

NEXORA CARE FLOW

Appointment Records – Exploratory Data Analysis (EDA)

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1. EDA Overview

The Exploratory Data Analysis (EDA) was conducted on the cleaned Nexora Care Flow Appointment Records dataset to understand historical appointment demand and identify patterns that could support the development of a future appointment-volume forecasting model. The analysis examined overall demand trends, clinic-level activity, day-of-week effects, seasonal patterns, holiday effects, appointment types, booking channels, booking lead times, and appointment outcomes.

The analysis provides an understanding of when, where, and how appointment demand changes, while also identifying variables that may be useful during the feature-engineering and forecasting stages.

2. Overall Appointment Trend

Monthly appointment volumes were calculated by resampling the `AppointmentDateTime` field. This was used to examine how appointment demand changed over the available period.

The monthly trend provides an initial view of fluctuations in demand and helps identify periods of relatively high or low appointment activity. Understanding these changes is important for Nexora Care Flow because significant variations in appointment volume can affect staffing requirements, room allocation, patient waiting times, and other downstream clinic operations.

3. Appointment Volume by Clinic

Appointment records were grouped by `ClinicID` and `ClinicName` to compare demand across the pilot clinics.

This analysis highlights differences in appointment volumes between clinics. Such differences are important because forecasting should ultimately provide clinic-level demand estimates, rather than assuming that all clinics experience the same appointment patterns.

The analysis also provides a basis for identifying clinics with particularly high or low levels of activity.

4. Day-of-Week Analysis

The `AppointmentDateTime` field was used to derive the day of the week for each appointment. Appointment volumes were then compared across Monday to Sunday.

This analysis investigates whether appointment demand follows a recurring weekly pattern. The clinic-level cross-tabulation further examines whether individual clinics have different busiest days.

Day-of-week effects are particularly relevant to forecasting because predictable variations in demand can help administrators plan staffing and room capacity ahead of time.

5. Seasonal Analysis

The existing SeasonFlag variable was examined to understand appointment volumes across the defined seasonal periods:

- Standard
- Flu Season
- Allergy Season

The analysis compares appointment activity across these categories and also examines seasonal volume by clinic.

This is relevant to the project because seasonal conditions may influence healthcare demand. The SeasonFlag variable can therefore provide a useful input for subsequent forecasting analysis, although conclusions about annual seasonality should be interpreted carefully when the available historical period is limited.

6. Holiday Effect

Holiday patterns were examined using the IsHoliday variable. Appointment records were classified as either Holiday or Non-Holiday, allowing appointment volumes to be compared between the two periods.

This analysis helps determine whether holidays are associated with changes in appointment demand. Holiday effects are potentially important for forecasting because clinics may experience lower or altered appointment activity around public holidays.

The HolidayName field was also cleaned so that records without a holiday were represented as "No Holiday", making the holiday information easier to interpret and use in subsequent analysis.

7. Appointment Type

The distribution of AppointmentType was analysed to determine which types of appointments account for the greatest proportion of recorded activity.

Understanding appointment types provides additional context about the composition of clinic demand. Different appointment types may have different operational requirements and may contribute differently to overall clinic workload.

8. Booking Channel

The BookingChannel variable was examined to identify how appointments entered the Nexora Care Flow platform.

Analysing booking channels provides insight into how patients access appointments and whether particular channels account for a substantial proportion of demand. This information can also help identify variables that may be useful when developing forecasting features.

9. Booking Lead Time

The distribution of `BookingLeadTimeDays` was examined using descriptive statistics, including measures such as the minimum, maximum, mean, median, and quartiles.

Booking lead time provides information about how far in advance appointments are scheduled. Comparing lead times across appointment types can help determine whether some types of appointments are typically planned further in advance than others.

This may be useful when considering how much forward visibility Nexora administrators have when planning staffing and clinic resources.

10. Appointment Status

The `status` variable was analysed to understand the distribution of appointment outcomes.

This provides an important data-quality and operational check, helping identify how appointments are represented in the dataset and whether different statuses occur frequently enough to warrant consideration during later modelling decisions.

EDA Summary

Overall, the EDA establishes an understanding of the main characteristics of appointment demand within the Nexora Care Flow dataset. The analysis examined overall trends, clinic-level differences, weekly patterns, seasonal effects, holidays, appointment types, booking channels, booking lead times, and appointment status.

The findings provide the foundation for the next project phase, Feature Engineering, where appointment records will be aggregated at the appropriate clinic and time level and transformed into modelling features such as calendar variables, lagged appointment volumes, and rolling averages.